This project is designed to give the experience of working with a particular topological space. Thematically, you arrive upon your space by way of some problem and it is your goal to understand what topological properties your space has. Through answering these questions you gain key insights into how your space behaves on its own and in the context of the problem from which it original arose.

With your assigned space, answer the following questions below in detail. Each question will be graded according to the 4 point scale used on the exams. The assigment is due Wednesday, July 29th through Canvas.

The Spaces:

A. Consider the quotient space of the closed interval $X = [0, 1]$ given by the partition $X^* = \{\{x\} : 0 < x < 1\} \cup \{\{0, 1\}\}$.

B. Let $\mathscr{L}_x$ be the line segment in $\mathbb{R}^2$ from the point $(0, 0)$ to $(1, x)$ and let $I = (\frac{1}{2}, 1] \times \{0\}$ where $(\frac{1}{2}, 1]$ is a half-open interval in $\mathbb{R}$. Define the set

$$X = \bigcup_{n=1}^{\infty} \mathscr{L}_{\frac{1}{n}} \cup I.$$

Consider X as a subspace of $\mathbb{R}^2$.

C. Let $X = [0, 1]$ and $\mathscr{S} = \{[0, b) : b > \frac{1}{2}\} \cup \{(a, 1] : a < \frac{1}{2}\}$. Consider the coarsest topology for which every element of $\mathscr{S}$ is open.

Questions:

1. Define a basis for your space. Create pictures of the basic open sets (Desmos is great for this).
2. Does your space have a basis consisting of countably many subsets? (Like we saw $\mathscr{B} = \{(p, q) : p, q \in \mathbb{Q}\}$ for $\mathbb{R}$)
3. Can you find a subspace of your space which is homeomorphic to $\mathbb{R}$?
4. Is your space Hausdorff?
5. Is your space connected? What are the components?
6. Is your space path connected? What are the path components?
7. Is your space compact?